# Towards Semantic Performance Measurement Systems for Supply Chain Management

Artturi Nurmi, Thierry Moyaux, and Valérie Botta-Genoulaz

Université de Lyon
INSA-Lyon, LIESP, 69621 Villeurbanne Cedex, France
{artturi.nurmi,thierry.moyaux,valerie.botta}@insa-lyon.fr

**Abstract.** The literature on Supply Chain Management (SCM) supports the integration of key business processes, including the performance management process, in order to increase the performance within and between the organisations. Nevertheless, the lack of proper shared performance metrics, frameworks and technology in order to design, implement and manage Performance Measurement Systems (PMSs), have been identified as problems for achieving integration. Semantics-based technologies, especially ontologies, have been suggested to address such problems. The purpose of this paper is to discuss how ontologies may be used to raise interoperability and shared understanding in inter-organisational PMSs from syntactic to semantic level by introducing the concept of Semantic PMS. We also suggest future research issues for approaching collaborative performance management problems.

**Keywords:** Performance Measurement Systems, Supply Chain Management, Ontology Engineering, SCOR, Business Process Management.

## 1 Motivation

A Supply Chain (SC) is “a set of three or more entities (organisations or individuals) directly involved in the upstream and downstream flows of products, services, finances, and/or information from a source to a customer” [1]. The goal of Supply Chain Management (SCM) is to increase the long-term performance of the individual entities and the SC as a whole via strategic collaboration, profit sharing, and by integrating and managing the flows of products, services, finances, and/or information between and within the SC entities [1-4]. Many problems exist in SCM due to the fact that the performance of any entity depends on the decisions of the others, and increasing the performance of the SC hence depends on the willingness and abilities of the individual entities [5]. Today, successful SCM requires the integration of demand- and supply-focused market knowledge and activities, and is based on high performing collaborative knowledge management processes, which provide aid in decision-making for the entities involved in the management of the SC [6].

One of the main problems for the goal of increasing the performance of an SC is the lack of proper SC performance measures/metrics which may result in decreased customer satisfaction, suboptimal performance, lost competitive advantage and conflicts within the SC [7]. Performance Measurement Systems (PMSs) also tend to stay

R. Meersman et al. (Eds.): OTM 2010 Workshops, LNCS 6428, pp. 239–248, 2010.

static although the need to keep measures aligned with strategy [8]. The integration of PMSs with other systems or management practices has also been identified as a problem [8]. Different frameworks for the measurement of performance have been proposed from which the Supply-Chain Operations Reference model (SCOR) is the most well-known. However, tools based on SCOR or other models that provide aid in decision making are scarce [9]. Problems also arise because of inconsistent terminologies and semantics caused by differences in business backgrounds, cultures and information systems [10]. For instance, an organisation may use "price" while its supplier prefers "value" to designate the same thing. Ontologies seem to have promising applications to address such issues. An ontology is a kind of formal model - "an engineering product consisting of a specific vocabulary used to describe a part of reality, plus a set of explicit assumptions regarding the intended meaning of that vocabulary" [11]. Few papers studying the applications of ontologies to SCM or BPM can be found, but the area is still relatively new and fragmented. Performance Management (PMgt) and PMSs have been almost completely neglected.

In this paper, we discuss the applications of ontologies from the point of view of Performance Management (PMgt) of SCs. For this purpose, a literature review is conducted on SCM, BPM, PMgt, Ontology Engineering, and Knowledge Management. Key articles are selected, analysed and synthesised in order to identify the state-of-the-art of PMgt and PMSs in a SC environment. The concept of semantic PMS is being discussed.

The main contribution of this paper is to present the idea of using ontology-based semantic technologies in the PMgt of SCs, particularly in the collaborative design, implementation and management of PMSs, which has not been discussed before. As another contribution, we shed light on some relevant problems related to current inter-organisational PMgt and PMS research, which allows us to provide direction for future research on semantic PMSs.

The paper is organized as follows. We first introduce the basics of SC PMgt and SCOR in Section 2. Section 3 summarises ontology engineering, Semantic Business Process Management (SBPM) and ontologies in SCM. In Section 4, we introduce our view for Performance Measurement Systems in SC environment. We conclude in the Section 5.

## 2 Supply Chain Performance Management

Performance measurement can be defined as "the process of quantifying the efficiency and effectiveness of action" where "action leads to performance" [12]. In an SC, efficiency can be explained as "how economically the SC can produce the product and/or service to match the demand" [13-15]. Effectiveness means "how well the SC is meeting the customer requirements on deliverables and customer service" [13-15]. From the point of view of Business Process Management (BPM), increasing the performance of SCs can be approached by "integration of key business processes" [16]. SCM is thus interested in the financial and non-financial aspects of Performance Management (PMgt), but proper performance measures and programs that address the individual organisations as well as the SC as a whole are needed [17]. In addition, frameworks for developing and designing such measures, and a shared understanding between the SC members has to be reached in order to efficiently work together [18]. Interoperable information systems are also needed for effective information sharing [19].

**Performance Measurement Systems**

A Performance Measurement System (PMS) is the core of a general management process often described as the *Performance Management (PMgt) process*, which defines how an organisation uses various systems to manage its performance [20]. According to a recent literature review, a PMS has only two necessary features: 1) to have *performance measures* and 2) to provide *supporting infrastructure* such as the information system facilitating the performance measures or also the personnel required to manage the PMS [21]. The only necessary role of the PMS is to measure performance and other roles might include strategy management, internal and external communication, influence behaviour, and learning and improvement [21]. Fig. 1 clarifies the role of collaborative performance measurement of SC partners. Collaborative performance measurement is the process that produces valuable information from the financial, non-financial, internal and external activities. Its' purpose is to aid the partner organisations in their decision making and taking actions aiming to increase the performance of the individual organisation and the performance of the SC as a whole. Performance measurement provides the feedback from day-to-day activities which form the data for monitoring the progress of operational, tactical and strategic decisions and to take corrective actions when needed. These corrective actions may be managerial actions, directed at changing the organisation such as: changing business processes, information systems, policies, instructions, personnel or other systems. Or these corrective actions might be organisational actions such as: producing products and services according to received orders, communicating with suppliers, customers and other parties by using the systems and tools provided by the managers. Collaborative measurement should be as efficient and effective as possible, but current internal PMSs are not capable for delivering the needed support. Many papers concerning performance measurement, and PMS design and implementation for SCs can be found, e.g., by Beamon [22], Lambert and Pohlen [7], Gunasekaran *et al.* [17], Gunasekaran and Ngai [19], Folan and Browne [23], Shepherd and Günter [8], and

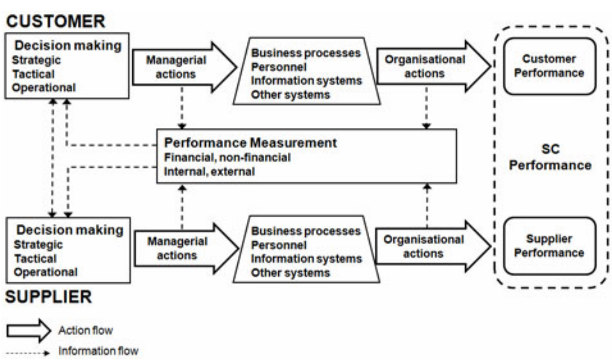


**Fig. 1.** Relationships Between Collaborative Performance Measurement and SC Performance

Jeschonowski *et al.* [24]. The common message of these papers is that SCM is still lacking proper frameworks, tools and technology in order to design, implement, and manage SC-wide PMSs. In fact, organisations are still struggling with implementing internal PMSs [25].

Findings from the literature support the need to integrate the PMgt process between SC members [9]. Different frameworks have been proposed, from which the Supply-Chain Operations Reference model (SCOR) is the most famous and may be considered as a standard [26].

**SCOR**

The SCOR model has been produced and maintained by the Supply Chain Council since 1996 and its purpose is to be used as a cross-functional business process reference model for SCM [27] . This model can be used for the modelling, re-design, and performance metrics design of SC processes. SCOR combines Business Process Reengineering (BPR), benchmarking, and best practices, and provides a cross-industry exhaustive hierarchical list of performance metrics with formal definitions. In SCOR, SCM is considered as the management of generic processes of Plan, Source, Make, Deliver and Return, and is intended to be used by all the members of the SC, from the suppliers' suppliers to the customers' customers. SCOR provides a common SC framework with standard terminology, and can also be used as a common model for evaluating, positioning, and implementing SC application softwares [28].

To get the best ouf of SCOR, like any other framework, it should be put into practice. Therefore, it has to be implemented in a manner that aids decision making aiming to solve business problems. Some papers describing tools using SCOR have been published, such as a computer-aided SC design tool [28] and IBM SmartSCOR [29], but in general, the lack of operational tools for SC PMgt has been acknowledged [9].

## 3 Ontology Engineering in SCM

Lately, ontologies have been proposed to approach these kinds of problems, since one of the advantages of ontologies is to capture complex relationships in order to define an SC [30]. An ontology is a kind of formal model, commonly referred to as "an explicit specification of a conceptualization", where conceptualization is "an abstract, simplified view of the world that we wish to represent for some purpose" [31]. An ontology is an engineering artefact built upon a specific vocabulary used to describe a certain reality, in addition to a set of explicit assumptions regarding the intended meaning of words in the vocabulary [32]. A long-term objective of research in ontologies is to produce libraries of reusable knowledge components and knowledge-based services that can be invoked over networks [33]. Ontologies have been suggested to benefit in 1) communication between people and organisations because ontologies enable shared understanding and communication between people with different viewpoints, 2) inter-operability between systems since ontologies can be used as inter-lingua between software components and 3) system engineering due to the fact that ontologies fulfil the requirements for specification, reliability and reusability for efficient knowledge management and transfer [34]. Ontologies form the basis for the Semantic Web technology, which is mostly composed of ontology languages, repositories, reasoners and query languages providing scalable methods and tools for machine-readable representation and knowledge management [35]. Semantic Web services are based on semantic web technologies allowing the representation, reasoning, discovery and composition of Web services in order to achieve a more complex service taking into account business rules and preconditions [36]. Because of the similarity of Web services/processes and business processes, semantic Web services can be used to form complex sequences of business activities allowing support for BPM. This area is referred to as Semantic Business Process Management (SBPM).

**Semantic Business Process Management (SBPM)**

BPM can be defined as "supporting business processes using methods, techniques and software to design, enact, control and analyze operational processes involving humans, organisations, applications, documents and other sources of information" [37]. The problem with current BPM approaches (without ontologies) is that it relies on human labour, which in turn is time-consuming, costly and erroneous [38]. Semantic Web services have been applied to the automation of BPM by adding semantic-based technologies to BPM in order to obtain SBPM [35]. The objective is to semantically enrich BPM with ontologies such that the manual work required in various steps, between putting the business perspective into the actual implementation of the information system, can be mechanized. This enables the process implementation to be more efficient and effective and, in addition, information sourcing from the semantically enriched space of processes becomes more effective as intelligent queries can be performed without the need for human intervention [35]. SBPM has been developed for some years now and resulted in promising theory and practice. From the point of view of PMgt, SBPM can also be used for performance monitoring based on machine reasoning [39]. However, problems are still to be solved such as how to harness the business people in the organisations to have the skills to manipulate ontologies because such people are not educated in this field yet.

**Ontologies for SCM**

Little research can be found on semantic SCM although SCM can be considered to be based on the integration of key business processes and thus is based on BPM. Chandra [40-43] has been quite active. First steps towards semantic SCM were probably taken in a paper introducing the idea of collaborative knowledge management based on ontology engineering to aid decision-making and SC integration [43]. From the point of view of PMgt, ontologies have also been suggested for developing inter-organisational information systems at the semantic level [44] and for automatic order monitoring [45]. An approach of using ontology engineering to capture distributed knowledge in the SC for building reusable and shareable simulation models based on SCOR have also been discussed [46], a tool for SC modelling and simulation based on an ontology and SCOR has been developed [47], and it has been applied in a real case for SC configuration and performance evaluation [48]. In general, approaching SC integration with ontologies could be beneficial, because there is evidence that collaborative ontology development and management can be used to support knowledge management [31, 35, 43], enterprise and SC modelling [34, 40, 41, 49], systems engineering [34, 42, 49] and BPM [35, 39, 50], which are all key a spects of successful SCM today and in the future. Efforts for combining SCOR with ontologies in the SC environment have been published, but we have not found any article discussing how ontologies might be used for the design, implementation and management of PMSs enabling better integration of SC-wide PMgt processes.

## 4 Semantic Performance Measurement Systems for SCM

Traditional approaches are not efficient or effective enough in providing support for the continuous management of knowledge, processes and systems for achieving optimal SC performance. Designing, implementing and managing a PMS is a challenge for an individual organisation for many reasons. In addition to the technical challenges, the sociocultural factors such as how the employees will accept changes in their rewarding system or does the management of the organisation actually use the PMS as it was intended to be used may cause problems [25]. A PMS project has a wide impact on the organisation and concerns all personnel – project failures are not unusual [25]. Even if an organisation has successfully implemented an internal PMS, developing and maintaining a functional PMS amongst several organisations is an even more complex issue, because intra-organisational PMSs are usually not designed to be connected together [23].

A supplier and a customer have to agree on how they will measure the performance of their shared inter-organisational processes, because "shared understanding results in integration, and integration leads in turn to organisational performance" [18]. For example, a supplier and a customer agree to measure the performance of the suppliers' "order-to-delivery process" which is responsible for delivering the products or services according to the order. The equivalent process of the customer is the "purchasing process" which is responsible for timing the purchases so that products or services are available according to the purchase order. For measuring the "order-to-delivery process", the supplier and the customer will use a measure "delivered on time", but this can mean different things. The supplier might understand that "delivered on time" means "shipped on time" while the customer might consider it as meaning "received on time". This would of course lead to a situation where the transportation time is not calculated and possible availability problems. After realising the miscomprehension, the supplier and customer would agree that the measure is as the customer has defined "received on time". This would not completely solve the problem as the supplier might understand that it means "first delivery received on time" while the customer would actually mean "delivered completely on time" if the delivery is carried out in several batches. Change is continuous, but having proper up-to-date performance measures that are defined and understood at the semantic level is a prerequisite for a high performing PMS. A flexible infrastructure for the PMS, capable of making fast and accurate changes at the semantic level, is needed.

Detecting semantic mismatches (such as described in the "order-to-delivery" and "purchasing process" example above) would aid the organisations in their PMS design, implementation and management. Ontology engineering can provide solutions for such modelling, communication and systems interoperability issues at the semantic level. Reference models such as SCOR could provide needed common framework, and definitions for processes and performance measures to support intra- and inter-organisational semantic PMS development efforts. Ontology-based semantic technologies could enable higher collaboration and integration within and between organisations, thus allowing seamless information flows and higher automation between organisations leading to increased overall performance. Based on the evidence discussed in this paper, we believe that ontology-based semantic technologies could provide solutions for the continuous management of knowledge, business processes, and various systems that are needed to be up-to-date in order to achieve a high performing PMS, thus leading to SC wide PMgt process integration.

## 5 Conclusion

In this article, we have identified some key problems for improving the performance of SCs. These problems are related to the design, implementation, and management of the Performance Management System (PMS), which is needed for the integration of the Performance Management (PMgt) process within and between the organisations. Problems arise from 1) the lack of proper shared performance metrics, frameworks and understanding, which are needed in the development of inter-organisational PMSs, 2) the lack of operational tools for the integration of the PMgt process, 3) the lack of efficiency and effectiveness of the traditional BPM approach to tackle the continuous need to transform business objectives into IT supported business processes, and 4) the need for a technology capable of approaching these problems at the semantic level.

We have discussed the applications of ontology-based technologies for BPM and SCM from the PMgt point of view. Our main contribution is presenting the idea of using ontologies for the benefit of the collaborative development of PMS – presenting the concept of semantic PMS - which has not been discussed before.

This research has also some limitations. Not all of the literature has been reviewed in this paper, but, instead, we have selected some key publications since the purpose of this paper is to provide an overview of the state-of-the-art of PMgt and PMSs in SC environment, and what kind of applications can be found based on ontologies.

Future research should be conducted on testing and validating the feasibility of using ontology-based semantic technologies for 1) reaching shared understanding in collaborative development of performance measures, and automatically detecting semantic mismatches, and 2) collaborative design, implementation and management of inter-organisational semantic PMSs, compared to traditional non-semantic approaches.

## References

1. Mentzer, J.T., DeWitt, W., Keebler, J.S., Min, S., Nix, N.W., Smith, C.D., Zacharia, Z.G.: Defining supply chain management. Journal of Business Logistics 22, 1–25 (2001)
2. Childerhouse, P., Towill, D.R.: Simplified material flow holds the key to supply chain integration. Omega 31, 17–27 (2003)
3. Fugate, B., Sahin F., Mentzer J.T.: Supply chain management coordination mechanisms. Journal of Business Logistics. 27. 129–161 (2006)
4. Towill, D.R.: The seamless supply chain - the predators strategic advantage. International Journal of the Techniques of Manufacturing 13, 37–56 (1997)

5. Swaminathan, J., Smith, S., Sadeh, N.: Modeling supply chain dynamics: a multiagent approach. Decision Sciences 29, 607–632 (1998)
6. Esper, T., Ellinger, A., Stank, T., Flint, D., Moon, M.: Demand and supply integration: a conceptual framework of value creation through knowledge management. Journal of the Academy of Marketing Science, 5–18 (2009)
7. Lambert, D.M., Pohlen, T.L.: Supply Chain Metrics. The International Journal of Logistics Management 12, 1–19 (2001)
8. Shepherd, C., Günter, H.: Measuring supply chain performance: current research and future directions. International Journal of Productivity and Performance Management 55, 242–258 (2006)
9. Forslund, H., Jonsson, P.: Obstacles to supply chain integration of the performance management process in buyer-supplier dyads. International Journal of Operations and Production Management 29, 77–95 (2009)
10. Ye, Y., Yang, D., Jiang, Z., Tong, L.: Ontology-based semantic models for supply chain management. The International Journal of Advanced Manufacturing Technology 37, 1250–1260 (2008)
11. Daconta, M.C., Smith, K.T.X., Obrst, L.J.: The Semantic Web: A Guide to the Future of XML, Web Services, and Knowledge Management. John Wiley & Sons Inc., Chichester (2003)
12. Neely, A., Gregory, M., Platts, K.: Performance measurement system design: A literature review and research agenda. International Journal of Operations & Production Management 25, 1228–1263 (1995)
13. Agarwal, A., Shankar, R., Tiwari, M.K.: Modeling the metrics of lean, agile and leagile supply chain: An ANP-based approach. European Journal of Operational Research 173, 211–225 (2006)
14. Jüttner, U., Christopher, M., Baker, S.: Demand chain management-integrating marketing and supply chain management. Industrial Marketing Management 36, 377–392 (2007)
15. Min, S., Mentzer, J., Ladd, R.: A market orientation in supply chain management. Journal of the Academy of Marketing Science 35, 507–522 (2007)
16. Lambert, D.M., Cooper, M.C.: Issues in Supply Chain Management. Industrial Marketing Management 29, 65–83 (2000)
17. Gunasekaran, A., Patel, C., McGaughey, R.E.: A framework for supply chain performance measurement. International Journal of Production Economics 87, 333–347 (2004)
18. Neely, A.: Business performance measurement: Unifying Theory and Integrating Practice. Cambridge University Press, Cambridge (2008)
19. Gunasekaran, A., Ngai, E.W.T.: Information systems in supply chain integration and management. European Journal of Operational Research 159, 269–295 (2004)
20. Bititci, U.S., Carrie, A.S., McDevitt, L.: Integrated performance measurement systems: a development guide. International Journal of Operations & Production Management 17, 522–534 (1997)
21. Franco-Santos, M., Kennerley, M., Micheli, P., Martinez, V., Mason, S., Marr, B., Gray, D., Neely, A.: Towards a definition of a business performance measurement system. International Journal of Operations & Production Management 27, 784–801 (2007)
22. Beamon, B.: Measuring supply chain performance. International Journal of Operations & Production Management 19, 275–292 (1999)
23. Folan, P., Browne, J.: A review of performance measurement: Towards performance management. Computers in Industry 56, 663–680 (2005)

24. Jeschonowski, D., Schmitz, J., Wallenburg, C., Weber, J.: Management control systems in logistics and supply chain management: a literature review. Logistics Research 1, 113–127 (2009)
25. Bititci, U., Mendibil, K., Nudurupati, S.S., Garengo, P.: Dynamics of performance measurement and organizational culture. International Journal of Operations and Production Management 26, 1325–1350 (2007)
26. Leukel, J., Kirn, S.: A Supply Chain Management Approach to Logistics Ontologies in Information Systems. In: Proceedings of the 11th International Conference on Business Information Systems (BIS 2008), pp. 95–105. Springer LNBIP, Innsbruck (2008)
27. Supply-Chain Council: Supply Chain Operations Reference Model. Version 9.0 (2008)
28. Huang, S.H., Sheoran, S.K., Keskar, H.: Computer-assisted supply chain configuration based on supply chain operations reference (SCOR) model. Computers & Industrial Engineering 48, 377–394 (2005)
29. Dong, J., Ding, H., Ren, C., Wang, W.: IBM SmartSCOR - a SCOR based supply chain transformation platform through simulation and optimization techniques. In: Proceedings of the 2006 Winter Simulation Conference, Winter Simulation Conference, Monterey, pp. 650–659 (2006)
30. Fayez, M.: An Automated Methodology for a Comprehensive Definition of the Supply Chain Using Generic Ontological Components. The Department of Industrial Engineering and Management Systems. University of Central Florida Orlando. p. 230 (2005)
31. Gruber, T.R.: Toward principles for the design of ontologies used for knowledge sharing. International Journal of Human-Computer Studies 43, 907–928 (1995)
32. Guarino, N.: Formal Ontology and Information systems. In: Proceedings of FOIS 1998, pp. 3–15. IOS Press, Amsterdam (1998)
33. Gruber, T.R.: A translation approach to portable ontology specifications. Knowledge Acquisition 5, 199–220 (1993)
34. Uschold, M., Gruninger, M.: Ontologies: principles, methods and applications. The Knowledge Engineering Review 11, 93–136 (1996)
35. Hepp, M., Leymann, F., Domingue, J., Wahler, A., Fensel, D.: Semantic Business Process Management: A Vision Towards Using Semantic Web Services for Business Process Management. In: Proceedings of the IEEE International Conference on e-Business Engineering, pp. 535–540. IEEE Computer Society, Washington (2005)
36. Sycara, K., Paolucci, M., Ankolekar, A., Srinivasan, N.: Automated discovery, interaction and composition of Semantic Web services. Web Semantics: Science, Services and Agents on the World Wide Web 1, 27–46 (2003)
37. van der Aalst, W., ter Hofstede, A., Weske, M.: Business Process Management: A Survey. In: van der Aalst, W.M.P., ter Hofstede, A.H.M., Weske, M. (eds.) BPM 2003. LNCS, vol. 2678, pp. 1–12. Springer, Heidelberg (2003)
38. Pedrinaci, C., Domingue, J., Brelage, C., van Lessen, T., Karastoyanova, D., Leymann, F.: Semantic Business Process Management: Scaling up the Management of Business Processes. In: Proceedings of the 2008 IEEE International Conference on Semantic Computing, pp. 546–553. IEEE Computer Society, Santa Clara (2008)
39. Wetzstein, B., Ma, Z., Leymann, F.: Towards Measuring Key Performance Indicators of Semantic Business Processes, In: Business Information Systems, pp. 227–238 (2008)
40. Chandra, C.: Supply Chain Workflow Modeling Using Ontologies. In: Collaborative Engineering, pp. 61–87. Springer, Boston (2008)
41. Chandra, C., Tumanyan, A.: Ontology Driven Knowledge Design and Development for Supply Chain Management. In: Proceedings 13th Annual Industrial Engineering Research Conference, Houston (2004)

42. Chandra, C., Tumanyan, A.: Organization and problem ontology for supply chain information support system. Data & Knowledge Engineering 61, 263–280 (2007)
43. Smirnov, A.V., Chandra, C.: Ontology-Based Knowledge Management for Cooperative Supply Chain Configuration. In: Proceedings of the American Association of Artificial Intelligence Spring Symposium, pp. 85–92. AAAI Press, Menlo Park (2000)
44. Hellingrath, B., Witthaut, M., Böhle, C., Brügger, S.: An Organizational Knowledge Ontology for Automotive Supply Chains. In: Holonic and Multi-Agent Systems for Manufacturing, pp. 37–46 (2009)
45. Zimmermann, R., Käs, S., Butscher, R., Bodendorf, F.: An Ontology for Agent-Based Monitoring of Fulfillment Processes. In: Ontologies for Agents: Theory and Experiences, pp. 323–345 (2005)
46. Fayez, M., Rabelo, L., Mollaghasemi, M.: Ontologies for supply chain simulation modeling. In: Proceedings of the 37th Conference on Winter Simulation, Winter Simulation, Conference, Orlando, pp. 2364–2370 (2005)
47. Cope, D., Fayez, M., Mollaghasemi, M., Kaylani, A.: Supply chain simulation modeling made easy: an innovative approach. In: Proceedings of the 39th conference on Winter simulation: 40 years! The best is yet to come, pp. 1887–1896. IEEE Press, Washington (2007)
48. Cope, D.: Automatic Generation of Supply Chain Simultion Models from SCOR Based Ontologies. Department of Industrial Engineering and Management Systems. University of Central Florida Florida (2008)
49. Uschold, M., Gruninger, M.: Ontologies and semantics for seamless connectivity. SIGMOD Rec. 33, 58–64 (2004)
50. Hepp, M., Roman, D.: An Ontology Framework for Semantic Business Process Management. In: Proceedings of Wirtschaftsinformatik, pp. 423–440. Universitätsverlag, Karlsruhe (2007)